

Objectives

Explain the basic structure and function of a neuron, including the cell body, dendrites, axon, and myelin sheath

Cells of the Nervous System

OBJECTIVES

Describe the Cells of the Nervous System, anatomy of a neuron and the roles of its main components: cell body, dendrites, and axon. types and function of Glia cells.

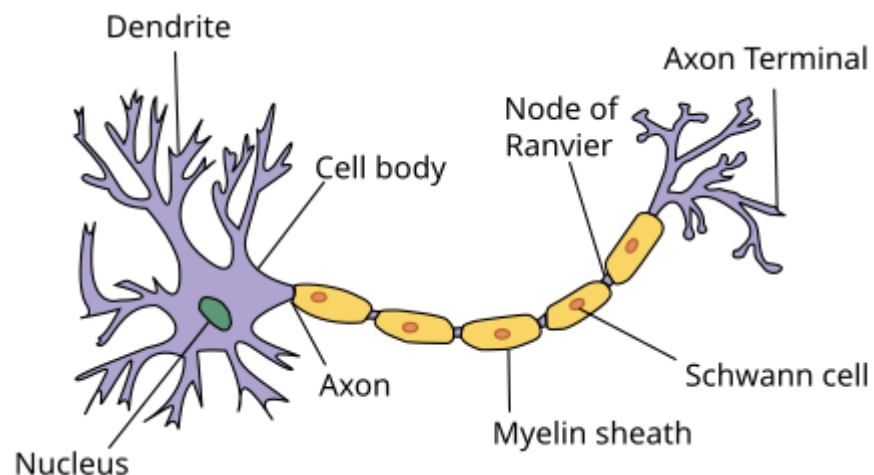
Cells of the Nervous System

The nervous system contains two main classes of cells: **neurons and glial cells**.

1. Neuron is the functional unit—that is, the smallest unit of a tissue that can carry out the function of that tissue. Neurons are “excitable cells” that communicate by transmitting electrical impulses. Excitable cells are defined as cells capable of producing large, rapid electrical signals called action potentials.

I. Neurons : Most neurons contain three main components: a cell body and two types of neural processes (or neurites) that extend from the cell body—the dendrite(s) and an axon. Figure 1

The cell body (also called soma or perikaryon)) contains the cell nucleus, endoplasmic reticulum, Golgi apparatus, and most of the free ribosomes. Mitochondria are located in the cell body, but also throughout the neuron. The cell body carries out most of the functions that other cells perform, such as protein synthesis and cellular metabolism. Although mature neurons retain their nuclei, they lose the ability to undergo cell division. Thus, in most areas of the nervous system, adults have all the neurons they will ever have.



Dendrites branch from the cell body and receive input from other neurons at specialized junctions called synapses. (Cell bodies themselves can also receive input at synapses.) The branching pattern of dendrites varies between different types of neurons, as shown for the star-shaped stellate cells and the highly branching Purkinje cells. The extent of branching is an indication of the number of synapses with the neuron, as the majority of synapses occur on the dendrites.

Neurons have another branch that comes off the cell body called an axon, or nerve fiber. Unlike a dendrite, whose function is to receive information, an axon sends information. Generally, a neuron has only one axon, but axons can branch, thereby sending signals to more than one destination. The branches of an axon are called collaterals, and as with dendrites, the extent of branching varies among neurons and is indicative of the amount of communication with other cells. The axons of many neurons acquire a myelin sheath, a protein–lipid complex that is wrapped around the axon. In the peripheral nervous system, myelin forms when a Schwann cell (a type of glia) wraps its membrane around an axon up to 100 times. The myelin sheath envelops the axon except at the nodes of Ranvier, periodic 1- μm gaps where the axon is unmyelinated.

Structural Classification of Neurons

Based on the number of processes that emanate from their cell body, neurons can be classified as unipolar, bipolar, pseudounipolar and multipolar , figure2.

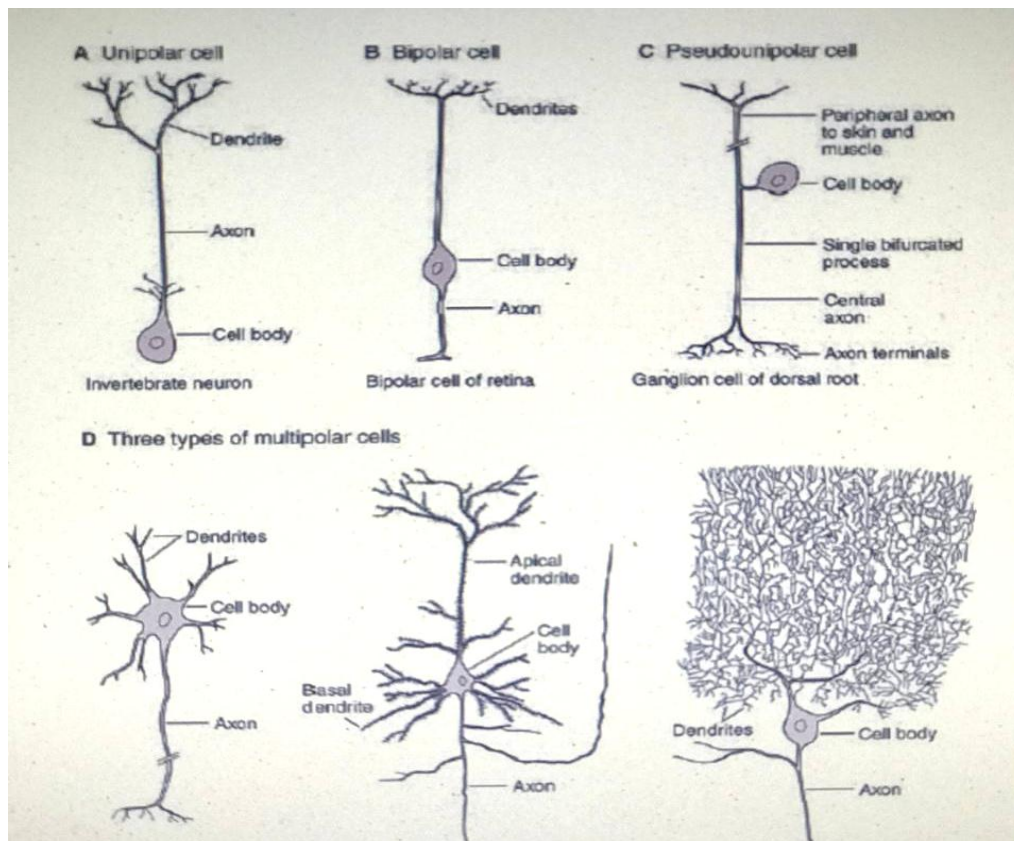


Figure 2 Structural Classification of Neurons

Functional Classification of Neurons

Three functional classes of neurons exist: efferent neurons, afferent neurons, and interneurons

1. afferent neurons is to transmit information from either sensory receptors or visceral receptors to the central nervous system for further processing
2. Efferent neurons transmit information from the central nervous system to effector organs

3. interneurons, which account for 99% of all neurons in the body. They are located entirely in the central nervous system. Interneurons perform all the functions of the central nervous system,

Another classification is according to conduction velocity and diameter of nerve fiber. Nerve signal conduction velocity increases with increased diameter and presence of electrical insulation (myelin sheath). Classes are:

Nerve type A α has the largest diameter and fastest conduction velocity e.g. somatic motor and proprioceptive nerve fibers.

Nerve type A β e.g. sensory fibers of fine touch and fine pressure.

Nerve type A γ e.g. motor fibers to muscle spindle.

Nerve type A δ e.g. sensory fibers of acute pain, crude touch and cold.

Nerve type B e.g. preganglionic autonomic nerve fibers.

Nerve type C has the smallest diameter unmyelinated fibers e.g. sensory fibers of chronic pain, heat, gross pressure and postganglionic sympathetic fibers.

II.Glia the second class of cell found in the nervous system, account for 90% of all cells in the nervous system. Their main functions include providing structural integrity to the nervous system (glia is Latin for “glue”) . and chemical and anatomical support that permits neurons to carry out their functions.

TYPES OF GLIA

.A.Microglia

B.Oligodendrocytes,

C.Schwann cells

D. Astrocytes

E. Ependymal cells

A. Microglia are part of the immune system; they are scavenger cells that resemble tissue macrophages and remove debris resulting from injury, infection

B. Oligodendrocytes are small with relatively few processes. Those in the white matter provide myelin, and those in the gray matter support neurons.

C. Schwann cells provide myelin to the peripheral nervous system. Each cell forms a segment of myelin sheath about 1 mm long; the sheath assumes its form as the inner tongue of the Schwann cell turns around the axon several times, wrapping in concentric layers. Intervals between segments of myelin are the nodes of Ranvier.

one oligodendrocyte sends out projections providing the myelin segments for many axons. In contrast, Schwann cells form myelin around axons in the peripheral nervous system, but each Schwann cell provides myelin for only one axon. Many oligodendrocytes or Schwann cells are needed to provide the myelin for a single axon. Figure 2

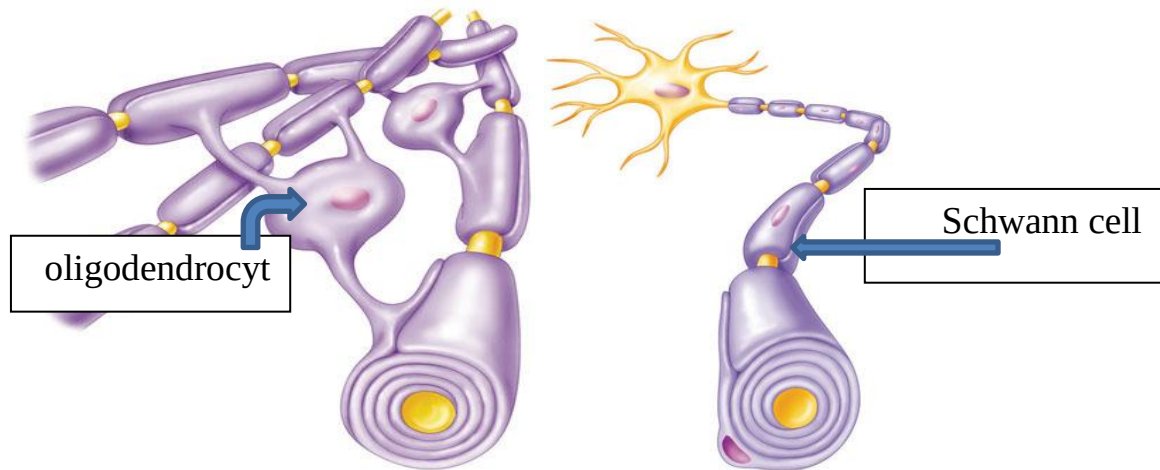


Figure 2 Oligodendrocyte and Schwann cells

D. Astrocytes are the most common glia in the CNS and are characterized by their starlike shape. They contact both capillaries and neurons and are thought to have a nutritive function. They are also involved in forming the blood-brain barrier.

E. Ependymal cells: They line the central cavities of brain and spinal cord

F. Satellite cells: *Type of glia in peripheral nervous system*. They surround cell bodies in PNS and have many of the same functions as astrocytes in CNS.

References

-Ganong and Guyton textbooks

-principle of human physiology 5th edition